



Audio deepfake detection based on simplifying graph convolution[☆]

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ABSTRACT

As low-artifact deepfake audio proliferates, the credibility of audio in personal, legal, and public communications is severely undermined, demanding reliable audio deepfake detection (ADD) solutions. However, previous graph-based approaches suffer from high computational overhead due to the nonlinearities between consecutive layers in graph attention networks. To address this limitation, we use Simplifying Graph Convolution (SGC) to optimize the original graph attention network (RawGAT-ST) by replacing iterative graph attention updates with linear feature aggregation. This modification reduces computational complexity while retaining the ability to capture critical spectro-temporal cues. Extensive experimental results show that the proposed detection method, an Optimizing Audio Deepfake Detection Network via Simplifying Graph Convolution (SGC-ADD), outperforms previous graph-based approaches with higher computational efficiency, by simplifying graph structure learning while preserving critical discriminative features.

1. Introduction

In the digital era, audio is indispensable in personal, legal, public communications and remote services [1], yet audio forgery technologies and deep learning methods have undermined its credibility via low-artifact deepfake audio, demanding reliable anti-forgery solutions [2–4]. Among existing approaches, passive audio deepfake detection (ADD) methods play a crucial role in identifying forged content, but current solutions face significant challenges in balancing accuracy and computational efficiency.

Recent research in deepfake audio detection focuses on graph-based and non-graph-based frameworks. Before the emergence of graph-based frameworks, researchers have explored various ADD methods with different feature domains, each with inherent limitations. First, traditional methods rely on handcrafted acoustic features combined with classic machine learning classifiers [5]. Although these methods are easy to implement, handcrafted features lack generalization ability and fail to capture the subtle artifacts of deepfake audio. Furthermore, Time-domain-based methods directly extract features from raw audio signals [6]; they have low computational complexity but are highly susceptible to noise interference and perform poorly on complex deepfake audio. Moreover, frequency-domain-based methods transform audio into the frequency domain to extract forged traces [7], but they discard temporal sequential information, leading to incomplete

feature representation. Finally, time–frequency-domain-based methods integrate temporal and frequency features [8], becoming the mainstream of ADD research. However, traditional time–frequency methods struggle to model long-range spectro-temporal dependencies, resulting in suboptimal detection performance.

Graph-based methods such as Raw graph attention network with spectro-temporal attention (RawGAT-ST) [9] and Audio anti-spoofing using integrated spectro-temporal graph attention networks (AASIST) [10] excel at capturing fine-grained spectro-temporal dependencies. However, RawGAT-ST's cascaded nonlinear attention layers lead to excessive computational costs (complexity $O(N^2)$), while AASIST further increases model complexity despite improved accuracy. Non-graph-based schemes [11] offer lower computational complexity but suffer from poor generalization to complex deepfake audio due to insufficient capture of long-range spectro-temporal dependencies. Additionally, SSL-antispoofing [12] leverages pre-trained models to enhance robustness but incurs higher inference latency, limiting real-time deployment.

In the realm of simplifying graph structure models, researchers have conducted extensive explorations, yielding several representative approaches with distinct characteristics and application scopes. Heterogeneous Graph Convolutional Network (HeteGCN) is tailored for heterogeneous graph data, which contains diverse types of nodes and

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edges [13]. This makes it excel in node classification tasks within citation networks, social networks, and text classification scenarios, particularly performing well in small-sample labeled training data settings. LightGCN, primarily utilized in recommendation systems, simplifies graph convolution by removing redundant nonlinear activation functions and feature transformations, focusing on the core graph convolution operation of feature smoothing [14]. GAAN (Graph Adaptive Attention Network) integrates an adaptive attention mechanism (AAM) and multi-head graph convolution (MHGC), enabling flexible capture of complex relationships between nodes [15]. It has demonstrated strong performance in text classification, user geolocation, and other tasks, with the adaptive attention mechanism allowing differential weighting of neighbor nodes to enhance feature representation capabilities. Simplifying Graph Convolution (SGC) stands out for its strong versatility and has been widely applied in node classification, text classification, relation extraction, and other cross-domain tasks. Simplifying Graph Attention Network (SimGAT) further optimizes the attention mechanism based on traditional graph attention networks (GAT) [16].

Simplified graph models are optimized traditional graph neural network variants with streamlined redundant components. HeteGCN simplifies critical node connection selection via heterogeneous information fusion and dynamic edge prediction but incurs computational overhead from complex feature transformations and nonlinear operations. LightGCN reduces graph convolution complexity by removing redundant activations and transformations to focus on core feature smoothing; it enables efficient training and inference via multi-layer embedding fusion without performance loss, though its simplification targets recommendation scenarios. GAAN simplifies complex node relationship capture using adaptive attention and multi-head convolution but retains high complexity from their combination. SGC collapses multi-layer graph convolution into a single linear transformation, drastically cutting complexity via logistic regression-based efficient training. Its linear mapping suits audio deepfake detection's efficiency-effectiveness balance for broad prospects. SimGAT streamlines attention coefficient calculation with a single attention head and target node independence, lowering costs compared with standard GAT while retaining neighborhood capture, yet its simplified attention limits feature representation for complex audio spectro-temporal dependencies.

Overall, these simplified graph models deliver strong results in their domains by optimizing efficiency and retaining key feature capture. However, their application in speech tasks especially audio deepfake detection remains scarce, leaving untapped potential for modeling audio structural dependencies. This gap highlights the need to adapt simplified graph convolution for audio deepfake detection, with SGC's linearization advantage making it ideal for balancing performance and efficiency.

To address these limitations, we propose an optimized deepfake audio detection framework, an Optimizing Audio Deepfake Detection Network via Simplifying Graph Convolution (SGC-ADD), that integrates Simplifying Graph Convolution (SGC) into the RawGAT-ST architecture. While SGC has been widely explored in node classification and text classification tasks, no prior work has applied linearized graph convolution to spectro-temporal graph modeling for speech anti-spoofing. This work is the first to introduce SGC into the audio deepfake detection domain. Different from existing methods that only focus on designing complex attention mechanisms, our core insight is that nonlinear graph updates are redundant for capturing spectro-temporal artifacts in deepfake audio. By collapsing multi-layer nonlinear GAT into a single linear aggregation, we reduce complexity while preserving discriminative cues. The core innovation lies in replacing cascaded nonlinear GAT layers with linear feature mapping, retaining spectro-temporal cue capture while reducing computational complexity.

Our main contribution is summarized as follows:

- We propose SGC-ADD, a novel deepfake-audio detection framework that optimizes RawGAT-ST by replacing its cascade of nonlinear layers with a single linear layer via simplifying graph

convolution, achieving a balance between detection accuracy and computational efficiency.

- We pioneer SGC-based linearization for audio deepfake detection, verifying its effectiveness across multiple mainstream ADD backbones. Extensive experiments on multiple datasets demonstrate the effectiveness of the proposed framework in accuracy and efficiency across diverse attack types and audio attributes.
- We further verify the generalization capability of the proposed SGC linearization strategy to mainstream audio anti-spoofing models (e.g., AASIST, SSL-DNN), providing insights for the design of lightweight and generalizable SGC-ADD models.

The remaining of this paper is organized as follows. Section 2 reviews related work in detail. Section 3 presents the SGC-ADD detection model, including architecture, linearization mechanism, and complexity analysis. Section 4 elaborates on the generalization applicability, adaptation mechanisms, and optimization principles of the proposed SGC-based linear optimization strategy for mainstream audio anti-spoofing models with diverse architectural characteristics. Section 5 conducts comprehensive experiments to benchmark detection accuracy and computational efficiency. Section 6 concludes with key findings, limitations, and future directions.

2. Related work and preliminaries

In this section, we comb through core related models that lay the foundation for our research to clarify the technical background and innovative positioning of the proposed model. These models mainly cover four categories, including Simplifying Graph Convolution(SGC) and audio forgery detection technology. The following subsections will elaborate on their structural features, core advantages, and respective roles in our research.

2.1. Simplifying Graph Convolution (SGC)

SGC is a lightweight Graph Convolutional Network (GCN) variant proposed to solve the problems of over-parameterization and high computational complexity of traditional GCNs [17]. It simplifies multi-layer graph convolution into a single-layer linear transformation followed by activation, removing inter-layer non-linearities and collapsing multi-layer operations into a unified linear projection. The key formula of SGC is

$$X_{\text{out}} = S^K X_{\text{in}} \Theta \quad (1)$$

where S is the normalized adjacency matrix, K is the receptive field size (controlling K -hop neighbor aggregation), X_{in} and X_{out} are the input and output feature matrices, respectively, and Θ is the merged weight matrix.

2.2. Audio forgery detection technology

2.2.1. Graph Attention Network (GAT)

As the core component of RawGAT-ST, GAT is a graph neural network (GNN) designed to address the limitation of traditional GCNs in handling node-specific importance [18]. First, unlike GCNs that assign fixed weights to neighboring nodes based on graph structure, GAT introduces an attention mechanism to adaptively learn the relevance between a target node and its neighbors. For each node, a feed-forward network first computes attention scores with its neighbors. These scores are then softmax-normalized so that more relevant neighbors dominate the feature update process, and multi-head attention is stacked to further stabilize the learned representations. Moreover, the flexibility of GAT in handling irregular graphs and its no reliance on precomputed Laplacians make it suitable for modeling audio frame nodes.

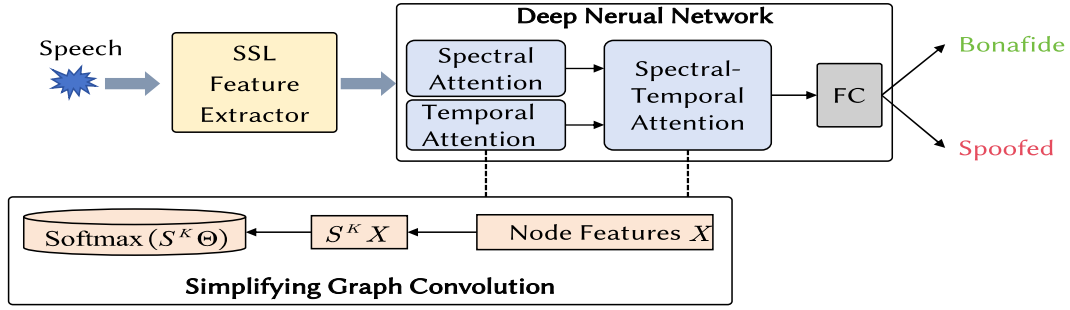


Fig. 1. Proposed audio deepfake detection method SGC-ADD. Spectral-temporal attention corresponds to the spectro-temporal fusion module. FC denotes the fully-connected network.

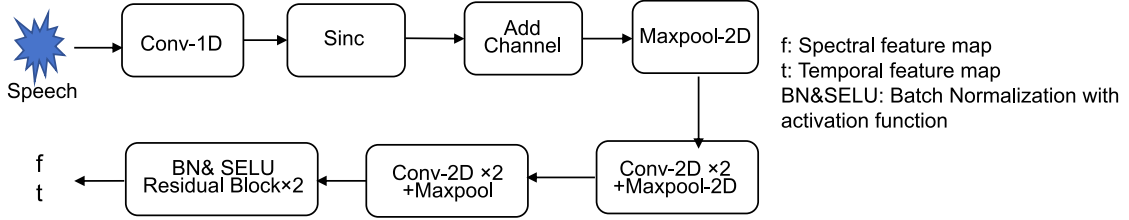


Fig. 2. Feature extractor.

2.2.2. Raw Graph Attention Network with Spectro-temporal Attention (RawGAT-ST)

RawGAT-ST is an end-to-end deepfake audio detection framework that leverages GAT and Sinc Layer to model spectral and temporal relationships [9]. Its core workflow and technical details include three parts:

(i) Feature Extraction with Sinc Layer: Raw audio waveforms are processed by a Sinc Layer—a learnable layer that models bandpass filters using sinc functions to capture fine-grained time–frequency details without manual feature engineering. The sinc function is defined in Eq. (2), where f_c is the learnable cutoff frequency. After Sinc Layer processing, a 2D residual backbone generates high-level feature maps $S \in \mathbb{R}^{C \times F \times T}$ (channels $\times$ frequency bins $\times$ temporal samples).

$$\text{sinc}(f_c, t) = \begin{cases} \frac{\sin(2\pi f_c t)}{2\pi f_c t} & t \neq 0 \\ 1 & t = 0 \end{cases} \quad (2)$$

(ii) Graph Construction: Parallel spectral and temporal attention branches construct domain-specific graphs. First, spectral graph G_f if formed by max-pooling S along the temporal dimension to extract frequency features $f \in \mathbb{R}^{C \times F}$, with nodes representing frequency bins and edges weighted by feature similarity. Moreover, temporal graph G_t if formed by max-pooling S along the frequency dimension to extract temporal features $t \in \mathbb{R}^{C \times T}$, with nodes representing temporal segments.

(iii) Graph Refinement with GAT: GAT layers refine G_f and G_t to model intra-domain relationships. The attention mechanism in GAT adaptively learns the relevance between nodes, prioritizing informative neighbors for feature aggregation.

(iv) Pooling, Fusion, and Classification: Graph pooling selects the most informative nodes from G'_f and G'_t , which are then aligned and concatenated into a spectro-temporal graph X_{ST} . A fusion GAT layer refines X_{ST} to capture inter-domain (spectral–temporal) relationships, and fully-connected layers output “bonafide” (genuine) or “spoof” (deepfake) classification probabilities.

A critical limitation of RawGAT-ST is the high computational complexity of the fusion GAT layer, which involves iterative attention calculations across all node pairs (complexity $O(N^2)$), hindering real-time applications 9.

3. Deepfake audio detection

3.1. Overall architecture

The proposed deepfake audio detection method integrates the RawGAT-ST model framework with the Simplifying Graph Convolution (SGC) method, which reduces model complexity while improving computational efficiency and detection robustness, and its overall structure is illustrated in Fig. 1. The detection process follows a sequential workflow that first extracts time–frequency features directly from raw audio waveforms using a learnable sinc layer. After feature extraction, spectral and temporal attention branches are employed to construct frequency and segment graphs respectively, and a graph-pooling stage is then applied to retain the most discriminative nodes from these two types of graphs to enhance feature discriminability. Next, the two graphs are first processed with an affine transformation to unify feature dimensions and resolve the node count mismatch, then aligned and concatenated to form a unified spectro-temporal graph, and an SGC layer is utilized to linearly aggregate neighborhood information within this fused graph to capture both intra-domain and inter-domain relational cues. Finally, two fully-connected layers are used to output the probability distribution of the audio belonging to “bonafide” (genuine) or “spoof” (deepfake) classes, thereby achieving end-to-end deepfake audio detection.

3.2. Feature extraction

The feature extraction process is demonstrated in Fig. 2, including Sinc Layer, Spectral Attention Block and Temporal Attention Block.

3.2.1. Sinc layer

The sinc layer serves as the initial component for raw-waveform feature extraction, and to obtain a time–frequency layout, an extra channel axis is appended to its output. This resulting tensor is then fed into a 2D residual backbone, which generates high-level feature representations denoted as $S \in \mathbb{R}^{C \times F \times T}$ where C , F , and T respectively index the number of channels, frequency bins, and temporal samples. The raw waveform feature extraction stage adopts the core Sinc layer logic in Section 2 (with the sinc function defined in Eq. (2)) to capture fine-grained time–frequency details. The topology of the residual backbone

is consistent with that described in the reference paper, with the key difference being that all convolutions employed here are 2D rather than 1D. Within each residual block, batch normalization is performed first, followed by the application of SeLU activations, a 2D convolutional layer, and a max-pooling step that reduces the data resolution to optimize subsequent feature processing. To capture domain-specific cues essential for deepfake detection, two parallel branches (spectral attention and temporal attention) are designed, and these branches first subject the high-level feature S to dimension-wise max-pooling on absolute values. This operation ensures that information carried by negative amplitudes, which may contain critical discriminative details, is not discarded.

3.2.2. Spectral attention block

The spectral attention block focuses on capturing cross-frequency correlations, and this is achieved by collapsing temporal information within the high-level feature S . Specifically, max-pooling is applied to the absolute values of S along the temporal dimension T , which yields a spectral feature map $f \in \mathbb{R}^{C \times F}$. The mathematical expression for this process is given by

$$f = \max_T(\text{abs}(S)) \quad (3)$$

where $\max_T(\cdot)$ represents the max-pooling operation performed over the temporal dimension, and $\text{abs}(\cdot)$ ensures that meaningful information contained in negative-valued data is preserved instead of being discarded, which is crucial for accurately capturing cross-frequency relational cues.

3.2.3. Temporal attention block

The temporal attention block emphasizes the capture of temporal continuity by collapsing spectral information in the high-level feature S . To achieve this, max-pooling is applied to the absolute values of S along the frequency dimension F , resulting in a temporal feature map $t \in \mathbb{R}^{C \times T}$. The corresponding mathematical formula is

$$t = \max_F(\text{abs}(S)) \quad (4)$$

where $\max_F(\cdot)$ denotes the max-pooling operation conducted over the frequency dimension. This process effectively isolates temporal patterns that are critical for deepfake audio detection, such as speech rhythm, which often exhibits distinct characteristics between genuine and deepfake audio.

3.3. Graph construction & GAT layer processing

To effectively capture and enhance the relational information within spectral and temporal features, a two-stage processing framework is designed for feature refinement. The first stage involves constructing graphs from the transposed spectral or temporal feature maps, which enables the modeling of relational cues between frequency bins (for spectral features) or temporal segments (for temporal features). The second stage employs separate Graph Attention Network (GAT) layers to refine these constructed graphs, which strengthens the discriminative features contained within the graphs and improves the subsequent detection performance. Fig. 3 presents the Graph Construction & GAT Layer Processing.

3.3.1. Graph construction

Graph construction consists of two parts, namely the construction of the spectral graph and the temporal graph. The spectral graph G_f is derived from the transpose of the spectral feature map f , i.e., $f^T \in \mathbb{R}^{F \times C}$, where each node in the graph represents a frequency bin. The number of nodes N_f is set to 23, which matches the number of spectral bins in the RawGAT-ST model, and the initial feature dimension of each node is 64, which is obtained through linear projection of the C -dimensional features of the spectral feature map. Thus, the spectral graph G_f has a dimension of $\mathbb{R}^{N_f \times d} = \mathbb{R}^{23 \times 64}$. The temporal graph G_t is derived from

the transpose of the temporal feature map t , i.e., $t^T \in \mathbb{R}^{T \times C}$, where each node corresponds to a temporal segment. The number of nodes N_t is 29, consistent with the number of temporal segments in the RawGAT-ST model, and the node feature dimension is also 64. As a result, the temporal graph G_t has a dimension of $\mathbb{R}^{N_t \times d} = \mathbb{R}^{29 \times 64}$.

3.3.2. GAT layer for graph refinement

The single-domain graph refinement stage adopts the core GAT aggregation logic in Section 2 to model intra-domain node dependencies. Each constructed graph (G_f and G_t) is processed by a GAT layer, which adaptively learns attention weights between nodes to prioritize informative neighbors during feature aggregation. This refinement process reduces the node feature dimension from $d = 64$ to $d' = 32$ while enhancing discriminative capabilities. After GAT layer processing, the spectral graph G_f is refined to $G'_f \in \mathbb{R}^{23 \times 32}$ and the temporal graph G_t is refined to $G'_t \in \mathbb{R}^{29 \times 32}$.

3.4. Graph pooling

All three GAT blocks (spectral, temporal, and spectro-temporal blocks) incorporate a graph pooling layer based on Graph U-Net, and the primary function of this layer is to select the most informative nodes from the refined graphs to further enhance feature discriminability [19]. The pooling process in Fig. 3 relies on a learnable projection vector $q \in \mathbb{R}^{d' \times 1}$, and its specific implementation involves three key steps. First, for each node n in G'_f (or G'_t) with a feature vector $X_n \in \mathbb{R}^{1 \times d'}$, the projection score y_n is computed through the dot product of X_n and the projection vector q . The mathematical expression for this calculation is

$$y_n = X_n \cdot q \quad (5)$$

where y_n measures the “informative value” of node n , and a higher score indicates that the node contains more critical features for deepfake audio detection. Second, the top- k nodes are selected based on the computed projection scores y_n , where k is a hyperparameter. For the refined spectral graph G'_f , k is set to 14, and for the refined temporal graph G'_t , k is set to 23. Let i denote the indices of these top- k nodes; the retained node features and their corresponding scores are then denoted as $X_{n,i} \in \mathbb{R}^{k \times d'}$ and $y_{n,i} \in \mathbb{R}^{k \times 1}$ respectively. Third, the retained node features are weighted by the sigmoid of their projection scores to emphasize the importance of high-value nodes, and this weighting process yields the pooled graph feature matrix. The formula for generating this matrix is

$$X' = X_{n,i} \cdot \text{sigmoid}(y_{n,i}) \quad (6)$$

where after the completion of the graph pooling process, the final pooled graphs are obtained: the spectral pooled graph $G'_{f,pooled} \in \mathbb{R}^{14 \times 32}$ derived from G'_f and the temporal pooled graph $G'_{t,pooled} \in \mathbb{R}^{23 \times 32}$ derived from G'_t .

3.5. Spectro-temporal fusion module

To capture the cross-domain relational cues between spectral and temporal features that are critical for tasks such as deepfake audio detection, a spectro-temporal fusion module is designed, and its structure is consistent with the aforementioned spectral and temporal feature processing modules. The operation of this module first addresses the dimension mismatch between the pooled spectral and temporal graphs to enable effective fusion, then fuses the features of the two graphs, and finally refines the fused representation using an attention-based layer to model the inter-domain dependencies between spectral and temporal features.

3.5.1. Feature alignment for cross-domain fusion

The pooled spectral graph (with 14 nodes and 32-dimensional features) and the pooled temporal graph (with 23 nodes and 32-dimensional features) have mismatched node counts, making direct

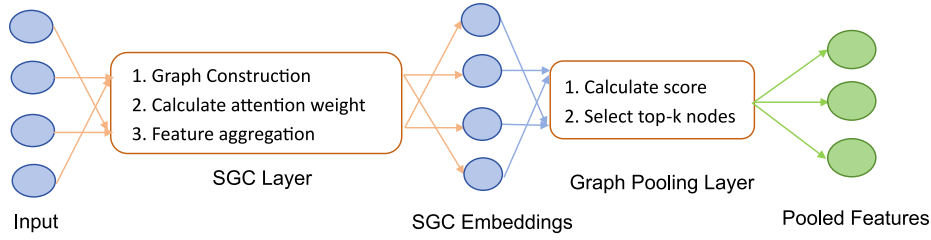


Fig. 3. Spectral & temporal attention.

fusion infeasible. To resolve this issue, an affine transformation is first applied to unify their feature dimensions to 64, matching the initial feature dimension of single-domain graphs, laying the foundation for subsequent cross-domain concatenation. The same affine transformation is applied to both the pooled spectral and temporal graphs to ensure feature dimension consistency. After alignment, the two feature matrices are concatenated along the node dimension to form the initial spectro-temporal fusion feature matrix, which contains 37 nodes (14 spectral + 23 temporal) with 64-dimensional features each. Each node in this matrix represents either a frequency bin or a temporal segment, encoding both domain-specific information from spectral and temporal domains and potential cross-domain relational information.

3.5.2. Spectro-temporal feature refinement

Similar to the single-domain feature refinement layers, the fusion refinement layer processes the initial spectro-temporal fusion feature matrix by referencing adaptive attention weighting mechanisms. This layer models both intra-domain dependencies within spectral or temporal nodes and inter-domain dependencies between spectral and temporal nodes via adaptive attention weighting. The refined spectro-temporal feature matrix is obtained by reducing the input feature dimension from 64 to 32, which balances representational capability and computational efficiency while integrating comprehensive cross-domain relational information.

3.6. SGC-based optimization

To address the high computational overhead associated with the iterative GAT layers in the RawGAT-ST model, the spectro-temporal GAT block is optimized using the SGC method. SGC is a linearized graph convolution technique that retains the receptive field expansion capability of Graph Convolutional Networks (GCNs) while removing redundant nonlinearities, and this optimization lays the foundation for reducing the computational costs of the overall detection model.

3.6.1. SGC linearization logic

Traditional K -layer GCNs rely on repeated neighborhood averaging operations using a normalized adjacency matrix S and layer-wise nonlinearities, which contribute to high computational complexity. SGC simplifies this architecture through three key improvements, consistent with the core logic of SGC in Section 2: (i) Removing inter-layer nonlinearities (retaining only the final classification softmax); (ii) Collapsing K -layer adjacency matrix iterations into S^K to expand the receptive field to K -hop neighbors; (iii) Merging K layer-wise weight matrices into a single matrix Θ to reduce parameters and computations. The mathematical expression for the merged weight matrix Θ is

$$\Theta = \Theta(1)\Theta(2) \dots \Theta(K) \quad (7)$$

where $\Theta(i)$ represents the weight matrix of the i th GCN layer in the traditional K -layer GCN structure.

3.6.2. SGC classification

For the fused spectro-temporal graph feature matrix $X_{ST} \in \mathbb{R}^{N_{ST} \times 64}$ — a matrix formed by concatenating the relevant fused graphs where

$N_{ST} = 14 + 23 = 37$ (the total number of nodes) — the output of the SGC layer in terms of detection probability is denoted as $\hat{Y}_{SGC}$, and its calculation follows the SGC core formula in Section 2 (refer to Eq. (1)):

$$\hat{Y}_{SGC} = \text{softmax}(S^K \cdot X_{ST} \cdot \Theta) \quad (8)$$

where $S \in \mathbb{R}^{N_{ST} \times N_{ST}}$ denotes the normalized adjacency matrix of X_{ST} , with edges constructed based on node feature correlation. S^K represents the K th power of S (empirically $K = 2$ for optimal receptive field), and $\Theta \in \mathbb{R}^{64 \times 2}$ is the merged weight matrix (output dimension = 2 for binary classification). This method replaces the iterative attention updates of RawGAT-ST, reducing computational complexity while retaining critical spectro-temporal cue capture capability.

3.7. Classification and detection output

This module serves as the final decision-making component of the deepfake audio detection model, and its primary function is to convert the learned graph-level features into clear binary detection results that distinguish between genuine and deepfake audio. The specific implementation of this module consists of two main parts, fully connected layers and detection output. The fully connected layers are responsible for mapping the graph-level feature F to a detection result, and two fully connected (FC) layers are employed for this purpose. The first FC layer uses a ReLU activation function, and its role is to further compress the dimension of the graph-level feature, which helps in reducing redundant information and enhancing the discriminative power of the feature. The second FC layer uses a softmax activation function, and it outputs the probability distribution of the input audio belonging to the “bonafide” class or the “spoofed” class—this probability distribution provides a quantitative measure of the model’s confidence in classifying the audio. The detection output step determines the final classification result based on the probability distribution generated by the second FC layer. The class with the higher probability is selected as the final detection result. For instance, if the probability of the audio being classified as “spoofed” is 0.85, the model will judge the audio as a deepfake audio clip; conversely, if the probability of “bonafide” is higher, the audio is classified as genuine. This binary decision-making process ensures that the model provides clear and interpretable detection results for practical application.

4. Generalization analysis

The linear optimization strategy based on Simplifying Graph Convolution (SGC) proposed in this study is not limited to the RawGAT-ST framework, but can be generalized to mainstream audio anti-spoofing models with different architectural characteristics. This section elaborates on the generalization applicability of the SGC linearization logic to two typical models (AASIST and SSL-DNN), focusing on the adaptation mechanism and optimization principles of the strategy for different model structures.

4.1. Generalization to AASIST

AASIST (Audio anti-spoofing using integrated spectro-temporal graph attention networks) is a representative graph attention-based audio anti-spoofing model, characterized by integrating multi-scale

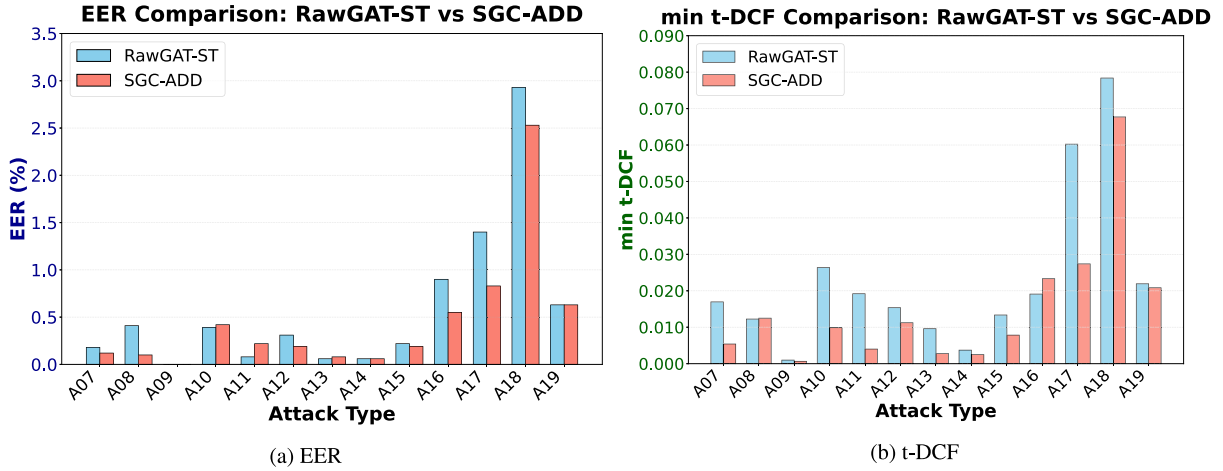


Fig. 4. Comparison of EER and t-DCF for SGC-ADD and RawGAT-ST across various audio forgery types.

spectro-temporal attention mechanisms to capture fine-grained dependency relationships in audio signals. Its core limitation lies in the cascaded multi-layer nonlinear attention computation process, which introduces redundant parameter updates and high computational overhead while enhancing detection robustness. The SGC linearization strategy can be seamlessly generalized to the AASIST model by replacing the multi-layer iterative spectro-temporal graph attention module in AASIST with a single-step SGC linear aggregation layer. This adaptation retains AASIST's inherent advantage of multi-scale spectro-temporal feature capture — by preserving the K-hop neighbor receptive field expansion capability of SGC (K is set according to AASIST's original attention layer depth) — while eliminating the nonlinear activation and repeated weight update between layers. The core optimization logic is to collapse the multi-layer adjacency matrix iteration and weight matrix of AASIST's graph attention module into a unified linear transformation, thereby reducing the computational redundancy caused by iterative attention scoring and softmax normalization between nodes.

4.2. Generalization to SSL-DNN

SSL-DNN (Self-Supervised Learning based DNN) is a typical self-supervised audio anti-spoofing model that leverages pre-trained self-supervised feature extractors to enhance the representation ability of audio features, especially for low-quality audio with complex noise interference. However, its deep fully-connected network structure in the spectro-temporal feature fusion stage leads to excessive model parameters and slow training convergence due to stacked nonlinear transformations. For SSL-DNN, the SGC generalization strategy is applied to the spectro-temporal feature fusion stage: the original multi-layer nonlinear transformation of DNN is replaced with SGC-based linear aggregation of spectral and temporal features. Specifically, the spectral and temporal features extracted by the self-supervised backbone are constructed into a unified spectro-temporal graph, and SGC is used to linearly aggregate the neighborhood information of graph nodes, replacing the nonlinear mapping of multiple fully-connected layers. This generalization approach retains the strong feature extraction capability of SSL-DNN's self-supervised pre-training while simplifying the feature fusion process — removing redundant nonlinear activation functions and reducing the number of model parameters — thus optimizing the training efficiency of the model without compromising its feature representation ability for complex audio signals.

5. Experiment

In this section, we conduct comprehensive experiments to verify the effectiveness, efficiency, and generalization capability of the proposed SGC-ADD method. Specifically, we compare its performance with

mainstream models, analyze computational complexity, and validate the generalization of the SGC linearization strategy to other audio anti-spoofing frameworks.

5.1. Settings

To evaluate the proposed method, we follow the practice of numerous studies [9,10,12] and adopt the logical access (LA) dataset from the ASVspoof 2019 corpus [20,21]. This dataset comprises six known audio attack types (A1–A6) used as the training set and thirteen unknown attack types (A07–A19) used as the test set. The logical access scenario of the ASVspoof 2019 dataset contains 19 spoofing attacks A01–A19. The training set includes six known attacks A01–A06, which are defined as follows:

- **A01:** TTS attack using HMM, RNN, VAE, and WaveNet vocoder.
- **A02:** TTS attack using HMM, RNN, VAE, and WORLD vocoder.
- **A03:** TTS attack using feed-forward acoustic model and WORLD vocoder.
- **A04:** TTS attack based on waveform concatenation.
- **A05:** VC attack using VAE framework and WORLD vocoder.
- **A06:** VC attack using GMM-UBM model and LPC synthesis.

The evaluation set consists of thirteen unseen attacks A07–A19 with various advanced synthesis and conversion mechanisms.

Notably, we also design generalization experiments to verify the applicability of the SGC linearization strategy to other mainstream audio anti-spoofing models (AASIST [10] and SSL-DNN [12]), with experimental settings consistent with the original models to ensure fairness.

To explore the detection performance under physical interaction scenarios, we also adopt the physical access (PA) dataset from the ASVspoof 2019 corpus:

- **AA:** Replay attack with high-grade playback device.
- **BB:** Replay attack with medium-grade playback device.
- **CC:** Replay attack with low-grade playback device.

These configurations cover diverse audio distortion degrees caused by different hardware devices, which is further categorized into low, medium and high signal quality levels corresponding to different SNR conditions in practical scenarios.

The core of the experiment focuses on comparing the performance between the original RawGAT-ST model and the improved model based on SGC denoted as SGC-ADD. As shown in Fig. 4, the last two columns represent the pooled Equal Error Rate (EER) [22] and pooled minimum

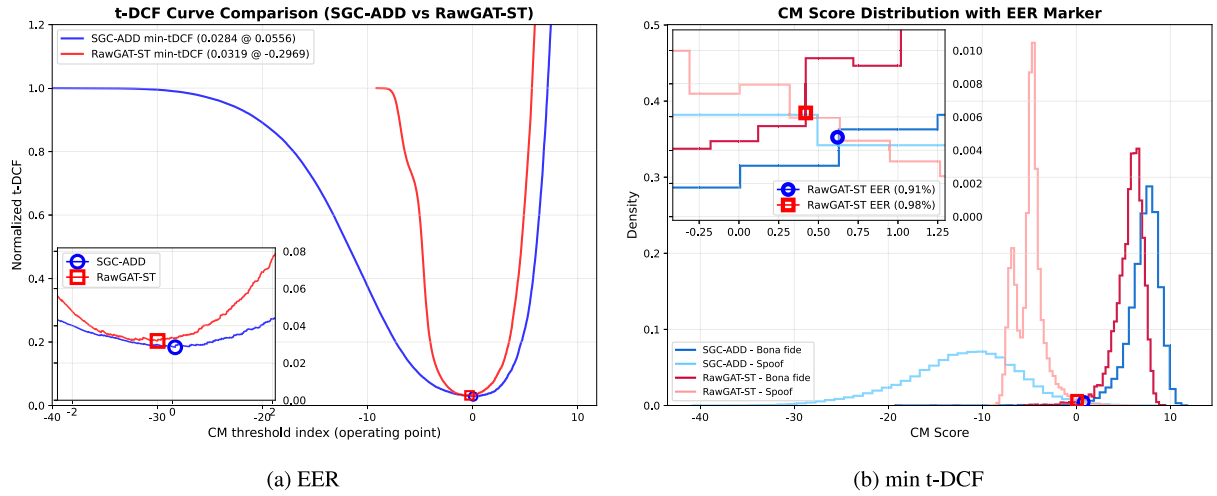


Fig. 5. The calculation of t-DCF and EER: (a) t-DCF curves of SGC-ADD (blue) and RawGAT-ST (red), with the minimum t-DCF values and corresponding countermeasure (CM) thresholds labeled. (b) Histograms of bonafide/spoof score distributions of the two models, with the score intersection corresponding to the Equal Error Rate (EER) marked.

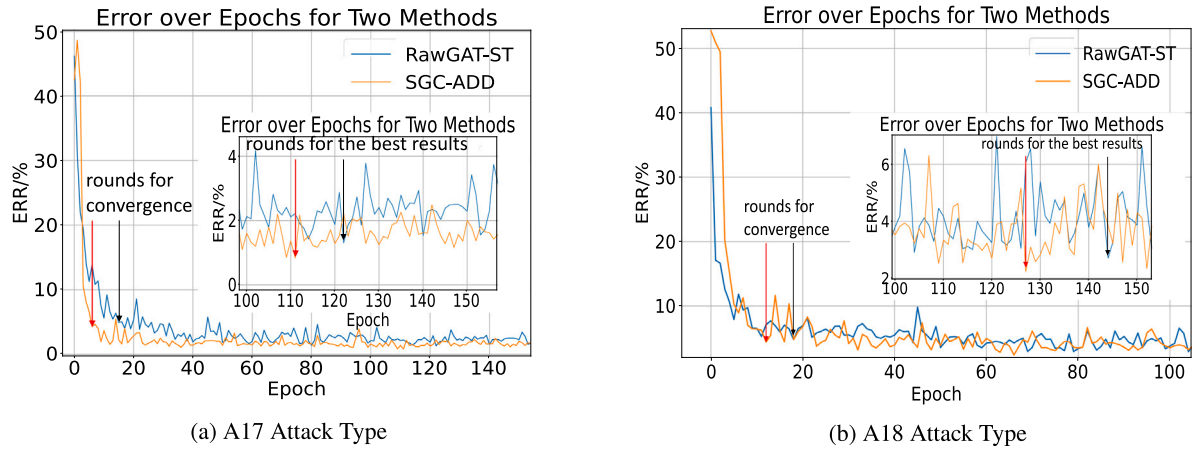


Fig. 6. Trend of EER with the number of training epochs.

Table 1

Comparison of computational complexity.

Model	Average time (ms)	Average increment (Mib)	FLOPs (M)
RawGAT-ST [9]	0.289	2.75	0.01
SGC-ADD [Ours]	0.274	2.58	0.01

Table 2

Comparison of average train time.

Model	Average time (min)	SGC-optimized (min)
RawGAT-ST [9]	17	15
SSL_DNN [12]	73	41
AASIST [10]	59	55

tandem Detection Cost Function (min t-DCF) [23,24] of each system, respectively. Specifically, min t-DCF reflects the anti-interference ability of the system with a lower value indicating stronger robustness, while EER reflects the detection efficiency of the system with a lower value indicating better balanced performance in accepting spoofed speech and rejecting bonafide speech. Fig. 6 further shows the variation trend of EER with the number of training epochs for the A17 attack type, and no additional complexity metrics for this model comparison are included in the current experimental design. Moreover, Table 1 shows the complexity of SGC-ADD and RawGAT-ST.

5.2. Result analysis

5.2.1. Experimental effectiveness

Experimental results show that SGC-ADD exhibits better performance in most attack types. As shown in Fig. 4, its average EER reduces

by 10.3% and min t-DCF reduces by 11.7%, both of which are lower than those of the original RawGAT-ST model. The performance gains are particularly pronounced under A08, A16, A17, and A18 attacks, where the proposed method outperforms the baseline by a clear margin. Fig. 4 further shows the t-DCF comparison of SGC-ADD and baseline models on A07–A19 attack types, which provides intuitive evidence for the superior performance of SGC-ADD. Significant gains on EER are observed on A08, A16, A17, and A18, where EER is reduced by more than 12% compared with the baseline, indicating that SGC linearization is highly effective for these typical synthetic and voice-conversion attacks. In contrast, the improvement is moderate on A07 and A10, where the baseline already yields very low EER, leaving limited room for further reduction.

Moreover, the overall t-DCF values are decreased across most attack types (11 out of 13), and a significant reduction of over 10% is

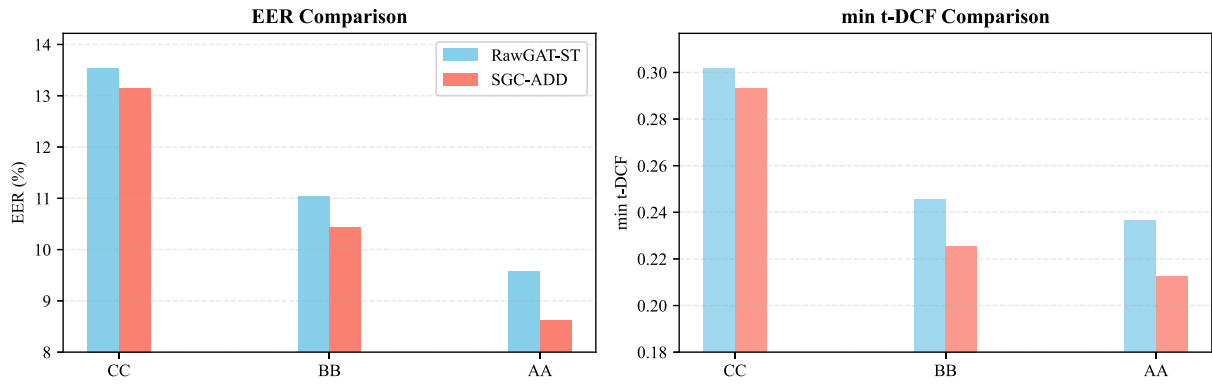


Fig. 7. Comparison of EER and t-DCF for SGC-ADD and RawGAT-ST across PA types.

achieved on A07, A10, A11, A13, A15, and A17 attacks, highlighting the method's stronger robustness against these typical spoofing attacks.

Fig. 7 shows that our method achieves consistent performance improvement against the baseline method across all physical access conditions. Compared with RawGAT-ST, our SGC-ADD reduces EER by 2.88%, 5.52%, and 9.93% and reduces min t-DCF by 2.85%, 8.07%, and 10.23% on CC, BB, and AA attacks, respectively. The optimization effect becomes more prominent under high-quality audio conditions, and still maintains stable detection capability under low-quality noisy environments, which fully verifies that our approach is well applicable to practical physical access anti-spoofing scenarios.

Additionally, Fig. 5 intuitively reflects the performance comparison of SGC-ADD and RawGAT-ST under different signal-to-noise ratio (SNR) conditions. It can be observed that SGC-ADD still maintains a lower EER than the baseline model even in low-SNR environments, which further supplements the performance verification of SGC-ADD under relevant experimental settings and confirms the effectiveness of the model in improving detection accuracy with strong environmental adaptability.

5.2.2. Computational complexity analysis

Under the same experimental conditions, the improved model requires fewer training epochs to converge. As reflected in Fig. 6, A17 and A18 are displayed in a single row, and the trend shows that the original RawGAT-ST takes more epochs to stabilize its EER, while SGC-ADD reaches a stable EER level faster. Specifically, for the A17 attack type, SGC-ADD converges at 6 epochs, with convergence epochs reduced by 53.8% and epochs to reach optimal performance reduced by 9.0% compared to RawGAT-ST. Similarly, for the A18 attack type, SGC-ADD also achieves significant improvement in convergence performance, with convergence epochs reduced and epochs to reach optimal performance compared to RawGAT-ST. These results confirm that SGC-ADD optimizes the model's training stability while maintaining or improving detection accuracy.

In terms of computational complexity, as shown in Table 1, the average inference time of SGC-ADD is 0.274 ms, which is 5.2% lower than RawGAT-ST's 0.289 ms; the average memory increment is 2.58 Mib, a 6.2% reduction compared to RawGAT-ST's 2.75 Mib. This verifies the advantage of SGC-ADD in reducing computational overhead. The reduction of computational overhead is achieved on the premise of improving EER and min t-DCF performances as discussed in Section 5.2.1. Meanwhile, this also verifies that SGC is a feasible and efficient simplification scheme for audio deepfake detection tasks.

5.2.3. Generalization capability verification

The generalization experiments verify that the SGC linearization strategy can be effectively applied to other mainstream audio anti-spoofing models. To rigorously evaluate the effectiveness of the proposed SGC-ADD, we compare it with three representative ADD models: RawGAT-ST (graph-based), AASIST (multi-scale graph attention),

and SSL-DNN (self-supervised learning-based). For AASIST, replacing its multi-layer spectro-temporal graph attention module with SGC retains the model's multi-scale feature capture advantage while reducing computational overhead; for SSL-DNN, applying SGC to the spectro-temporal feature fusion stage simplifies the multi-layer nonlinear transformation process without compromising the feature representation ability of the self-supervised pre-trained backbone. We further evaluate the generalization performance on unseen spoofing attacks. Additional data augmentation does not yield consistent improvements and may degrade detection accuracy, as it distorts subtle spectro-temporal cues critical for deepfake detection. Therefore, data augmentation is not incorporated into the final pipeline. The proposed method inherently supports generalization to unseen attacks, offering a streamlined solution for practical applications.

To further verify the generalization capability of SGC-ADD applying to different baseline models, a dedicated analysis of Figs. 8 and 9 is conducted. Specifically, Fig. 8 presents the EER and min t-DCF comparison curves between SGC-ADD and the baseline AASIST model on the attack types of the LA dataset. It can be observed that SGC-ADD maintains lower EER and t-DCF levels than the original AASIST model on most attack types (12 out of 13), which intuitively confirms the adaptability of the SGC linearization strategy to the AASIST model. Fig. 9 further shows the EER and t-DCF comparison curves between SGC-ADD and another baseline SSL-DNN model on the same dataset. The results indicate that SGC-ADD also achieves lower t-DCF values compared to the original SSL-DNN model on most attack types (9 out of 13). From the EER results, SGC-ADD achieves lower EER on most attack types (8 out of 13). These results fully demonstrate the strong generalization of the SGC linearization optimization strategy—even when switching to different baseline models, the proposed strategy can still effectively improve model performance. On the contrary, by simplifying redundant nonlinear operations, SGC reduces model complexity while adapting to different baseline models, further verifying the universal effectiveness of the proposed method on different mainstream audio anti-spoofing models. Specifically, compared with RawGAT-ST, AASIST, and SSL-DNN, our method reduces the average EER by 10.3%, 8.7%, and 9.2%, respectively, and achieves lower min t-DCF across most attack types.

As shown in Table 2, SGC-ADD reduces the average training time of RawGAT-ST from 17 min to 15 min, and also achieves significant training time reduction for SSL-DNN (from 73 min to 41 min) and AASIST (from 59 min to 55 min). The complexity advantage stems from the linearization optimization of SGC, which eliminates redundant nonlinear operations and iterative attention calculations in GAT layers, verifying the effectiveness of the proposed method in improving computational efficiency and its good generalization in different models.

SGC-ADD has minor limitations with marginal EER degradation on low-SNR samples and limited generalization to physical access attack scenarios. However, overall, the method achieves a balance between detection accuracy, computational efficiency, and generalization

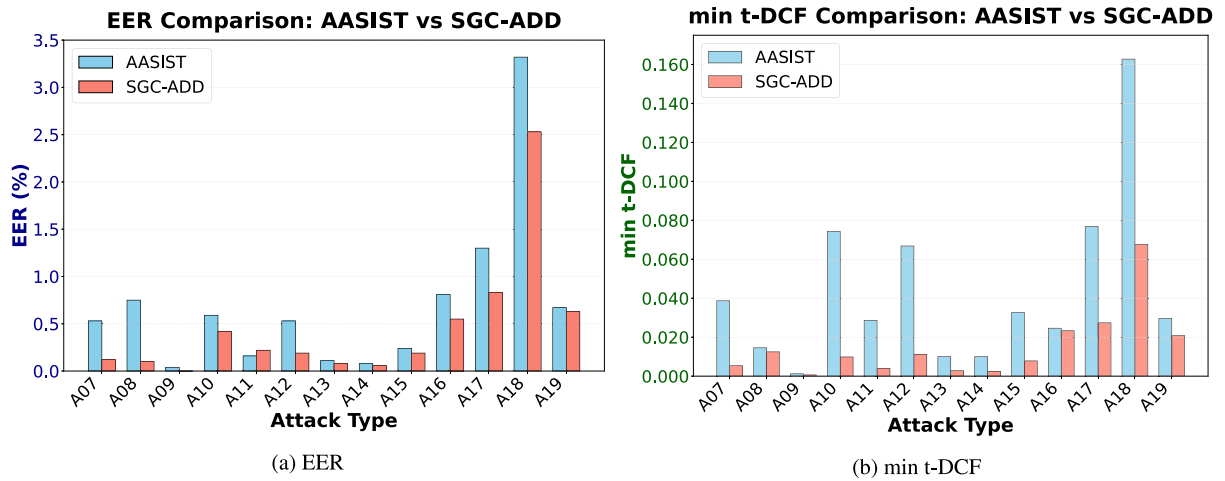


Fig. 8. Comparison of EER and t-DCF for SGC-ADD and AASIST across various audio forgery types.

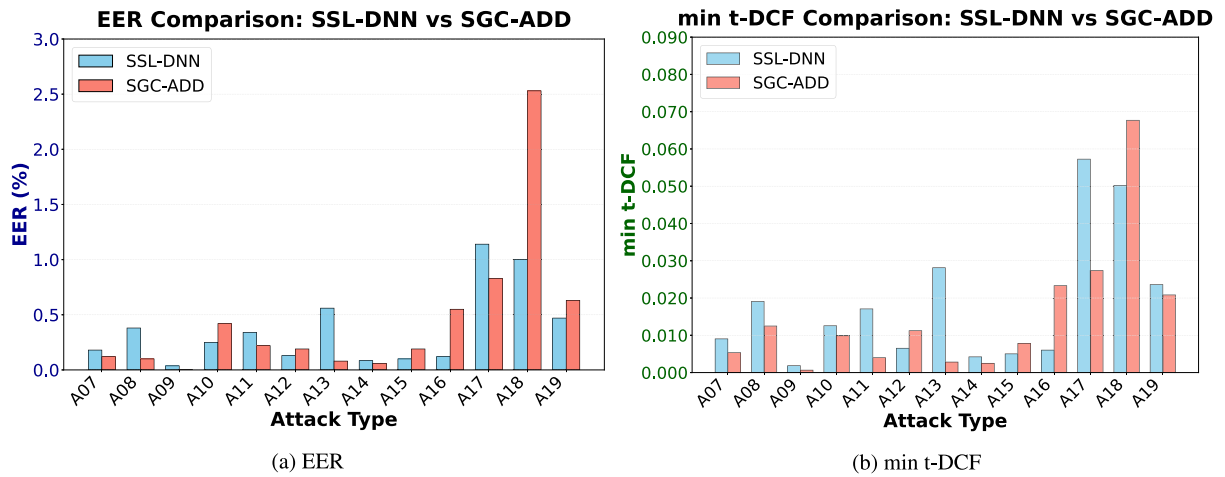


Fig. 9. Comparison of EER and t-DCF for SGC-ADD and SSL-DNN across various audio forgery types.

capability, providing a new idea for the design of lightweight and generalizable audio deepfake detection models.

The consistent improvement across different model architectures demonstrates that the SGC linearization strategy is generally applicable to mainstream audio anti-spoofing frameworks, and is not restricted to a specific network design.

6. Conclusion

In this paper, we address the high computational redundancy caused by successive nonlinear layers in traditional graph neural networks for audio deepfake detection. By introducing Simplifying Graph Convolution (SGC), we replace the iterative nonlinear attention updates in RawGAT-ST with a single linear feature aggregation step. This modification significantly reduces computational complexity while preserving the model's ability to capture critical spectro-temporal dependencies, enabling real-time scalability without compromising detection accuracy. Experimental results on the ASVspoof 2019 LA dataset verify that SGC-ADD outperforms the original RawGAT-ST model, with a reduction in average Equal Error Rate (EER) and minimum tandem Detection Cost Function (min t-DCF). Additionally, SGC-ADD exhibits faster convergence, reducing training epochs for specific attack types, which further confirms its efficiency advantage. Overall, this work advances practical anti-forgery capabilities for audio communications, bolstering audio credibility across personal, legal, and public sectors.

CRediT authorship contribution statement

Yufei Liu: Writing – original draft, Validation, Software, Methodology, Data curation, Conceptualization. **Zhangyao Song:** Writing – review & editing, Software, Conceptualization. **Tao Guo:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization. **Huihui Wu:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Tao Guo reports financial support was provided by National Natural Science Foundation of China. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

- [1] M.A. Nasr, W. El-Shafai, N. Abdel-Salam, E.S.M. El-Rabaie, A.S. El-Fishawy, F.E. Abd El-Samie, A comprehensive survey of audio forgery detection: challenges and novel trends, *J. Electr. Syst. Inf. Technol.* 12 (1) (2025) 30, <http://dx.doi.org/10.1186/s43067-025-00225-w>.
- [2] Y. Wang, R.J. Skerry-Ryan, D. Stanton, Y. Wu, R.J. Weiss, N. Jaitly, Z. Yang, Y. Xiao, Z. Chen, S. Bengio, Q.V. Le, Y. Agiomyriannakis, R. Clark, R.A. Saurous, Tacotron: Towards End-to-End speech synthesis, in: *Proc. Interspeech* 2017, 2017, pp. 4006–4010, <http://dx.doi.org/10.21437/INTERSPEECH.2017-1452>.
- [3] T.P. Doan, K. Hong, S. Jung, GAN discriminator based audio deepfake detection, in: *Proc. 2nd Workshop on Security Implications of Deepfakes and Cheapfakes, WDC*, 2023, pp. 29–32, URL <https://doi.org/10.1145/3595353.3595883>.
- [4] H. Gao, K. Qian, J. Ni, C. Gan, M.A. Hasegawa-Johnson, S. Chang, Y. Zhang, Speech Self-Supervised learning using diffusion model synthetic data, in: *Proc. 41st Int. Conf. on Machine Learning, ICML*, Vol. 235, 2024, pp. 14790–14810.
- [5] C. Bisogni, V. Loia, M. Nappi, C. Pero, Acoustic features analysis for explainable machine learning-based audio spoofing detection, *Comput. Vis. Image Underst.* 249 (2024) 104145, <http://dx.doi.org/10.1016/j.cviu.2024.104145>.
- [6] J. Li, H. Wang, F. Xie, X. Feng, J. Chen, J. Liu, J. Wang, Audio deepfake detection via a fuzzy Dual-Path Time-Frequency attention network, *Sensors* 25 (24) (2025) <http://dx.doi.org/10.3390/s25247608>.
- [7] C. Tan, Y. Zhao, S. Wei, G. Gu, P. Liu, Y. Wei, Frequency-Aware deepfake detection: Improving generalizability through frequency space learning, 2024, *ArXiv abs/2403.07240* URL <https://api.semanticscholar.org/CorpusID:268890333>.
- [8] B. Wang, Y. Tang, F. Wei, Z. Ba, K. Ren, FTDKD: Frequency-Time domain knowledge distillation for Low-Quality compressed audio deepfake detection, *IEEE/ACM Trans. Audio Speech Lang. Process.* 32 (2024) 4905–4918, <http://dx.doi.org/10.1109/TASLP.2024.3492796>.
- [9] H. Tak, J. Jung, J. Patino, et al., End-to-End Spectro-Temporal graph attention networks for speaker verification Anti-Spoofing and speech deepfake detection, in: *Proceedings of the ASvspoof Workshop*, 2021, pp. 1–5.
- [10] H. Tak, J. Jung, J. Yamagishi, AASIST: Audio Anti-Spoofing using integrated Spectro-Temporal graph attention networks, in: *Proceedings of the IEEE ICASSP*, 2022, pp. 6815–6819.
- [11] H. Tak, J. Patino, M. Todisco, A. Nautsch, N. Evans, A. Larcher, End-to-End anti-spoofing with RawNet2, in: *IEEE International Conference on Acoustics, Speech and Signal Processing, ICASSP*, 2021, pp. 6369–6373.
- [12] H. Tak, M. Todisco, X. Wang, et al., Automatic speaker verification spoofing and deepfake detection using wav2vec 2.0 and data augmentation, in: *The Speaker and Language Recognition Workshop*, 2022, pp. 1–6.
- [13] R. Ragesh, S. Sellamanickam, A. Iyer, R. Bairi, V. Lingam, HeteGCN: Heterogeneous Graph Convolutional Networks for Text Classification, New York, NY, USA, 2021, <http://dx.doi.org/10.1145/3437963.3441746>.
- [14] X. He, K. Deng, X. Wang, Y. Li, Y. Zhang, M. Wang, LightGCN: Simplifying and powering graph convolution network for recommendation, in: *Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2020, URL <https://api.semanticscholar.org/CorpusID:211043589>.
- [15] Z. Chen, Graph adaptive attention network with Cross-Entropy, *Entropy* 26 (7) (2024) URL <https://www.mdpi.com/1099-4300/26/7/576>.
- [16] H. Guo, R. Yan, Y. Feng, X. Gao, Z. Zhu, Simplifying graph attention networks with Source-Target separation, in: *ECAI 2020 - 24th European Conference on Artificial Intelligence*, 29 August–8 September 2020, Santiago de Compostela, Spain, August 29 - September 8, 2020 - Including 10th Conference on Prestigious Applications of Artificial Intelligence (PAIS 2020), 325, 2020, pp. 1166–1173, <http://dx.doi.org/10.3233/FAIA200215>.
- [17] F. Wu, A. Souza, T.Y. Zhang, C. Fifty, T. Yu, K. Weinberger, Simplifying graph convolutional networks, in: *Proceedings of the 36th International Conference on Machine Learning*, PMLR, 2019, pp. 6861–6871.
- [18] P. Velickovic, G. Cucurull, A. Casanova, A. Romero, P. Liò, Y. Bengio, Graph attention networks, in: *6th International Conference on Learning Representations, ICLR 2018*, Vancouver, BC, Canada, April 30 - May 3, 2018, Conference Track Proceedings, 2018, URL <https://openreview.net/forum?id=rJXMpikCZ>.
- [19] H. Gao, S. Ji, Graph U-Nets, in: *International Conference on Machine Learning*, 2019, pp. 2083–2092.
- [20] X. Wang, J. Yamagishi, M. Todisco, H. Delgado, A. Nautsch, N. Evans, M. Sahidullah, V. Vestman, T. Kinnunen, K.A. Lee, L. Juvela, P. Alku, Y.H. Peng, H.T. Hwang, Y. Tsao, H.M. Wang, S. Le Maguer, M. Becker, F. Henderson, R. Clark, Y. Zhang, Q. Wang, Y. Jia, K. Onuma, K. Mushika, T. Kaneda, Y. Jiang, L.J. Liu, Y.C. Wu, W.C. Huang, T. Toda, K. Tanaka, H. Kameoka, I. Steiner, D. Matrouf, J.F. Bonastre, A. Govender, S. Ronanki, J.X. Zhang, Z.H. Ling, ASvspoof 2019: A large-scale public database of synthesized, converted and replayed speech, *Comput. Speech Lang.* 64 (2020) 101114, <http://dx.doi.org/10.1016/j.csl.2020.101114>.
- [21] M. Todisco, X. Wang, V. Vestman, M. Sahidullah, H. Delgado, A. Nautsch, J. Yamagishi, N. Evans, T. Kinnunen, K.A. Lee, ASvspoof 2019: Future horizons in spoofed and fake audio detection, in: *Interspeech* 2019, 2019, pp. 1008–1012, <http://dx.doi.org/10.21437/Interspeech.2019-2249>.
- [22] N. Brümmer, E. de Villiers, The BOSARIS toolkit: Theory, algorithms and code for surviving the new DCF, 2013, *arXiv:1304.2865* URL <https://arxiv.org/abs/1304.2865>.
- [23] T. Kinnunen, K.A. Lee, H. Delgado, N. Evans, M. Todisco, M. Sahidullah, J. Yamagishi, D.A. Reynolds, T-DCF: a detection cost function for the tandem assessment of spoofing countermeasures and automatic speaker verification, in: *The Speaker and Language Recognition Workshop (Odyssey 2018)*, 2018, pp. 312–319, <http://dx.doi.org/10.21437/Odyssey.2018-44>.
- [24] T. Kinnunen, H. Delgado, N. Evans, K.A. Lee, V. Vestman, A. Nautsch, M. Todisco, X. Wang, M. Sahidullah, J. Yamagishi, D.A. Reynolds, Tandem assessment of spoofing countermeasures and automatic speaker verification: Fundamentals, *IEEE/ACM Trans. Audio Speech Lang. Process.* 28 (2020) 2195–2210, <http://dx.doi.org/10.1109/TASLP.2020.3009494>.